CS1 Week 12 Annotated

Time-delays, Designing Compensators, Non-Linearities

Attention! 🚨

- I will have to miss next week's exercise class on Friday, 12 December due to a review for my Focus Project…
- I'm planning something a bit more fun for the last week (Recap Kahoot?, short insight into my Focus Project if you are interested?)
- Let me know if you have any other suggestions for the last week!

Recap Last Week

Quiz 1

Which statements are true?

A. Commands and disturbances act at lower frequencies

B. Noise acts at lower frequencies

C. Commands and disturbances act at higher frequencies

D. Noise acts at higher frequencies

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A. Commands and disturbances act at lower frequencies

B. Noise acts at lower frequencies

C. Commands and disturbances act at higher frequencies

D. Noise acts at higher frequencies

Quiz 2

Which statements are true?

A. The sensitivity function $S(s)$ describes the impact of disturbances on the error

B. The sensitivity function $S(s)$ describes the impact of noise on the error

C. The complementary sensitivity function $T(s)$ describes the impact of noise on the error

D. $S(s) + T(s)$ is always equal to 1

Quiz 2

Which statements are true?

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B. The sensitivity function $S(s)$ describes the impact of noise on the error

C. The complementary sensitivity function $T(s)$ describes the impact of noise on the error

D. $S(s) + T(s)$ is always equal to 1

Quiz 3

What is the bandwidth of a system?

A. characterizes how fast the system can respond to changes in the reference signal

B. The bandwidth is the maximum $\boldsymbol{\omega}$ such that $|\boldsymbol{T(j\omega)}| > \mathbf{1/\sqrt{2}}$ $(\approx\ -\mathbf{3}\ \boldsymbol{dB})$

C. The bandwidth is often approximated by $\boldsymbol{\omega_c}$

D. Describes how fast your Inno Project robot will fail on qualification day

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Quiz 4

The maximum phase shift of a lead compensator occurs at $\omega = \sqrt{(ab)}$

$$\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$$

Question 42 *Choose the correct answer. (1 Point)*

You want to increase the phase at the crossover frequency $\omega_c = 1\,rad/s$ by $30°$ by using a lead-element $C(s) = \frac{s/a+1}{s/b+1}$. Which parameters a, b should you choose?

- A $a = 1, \quad b = 1$
- B $a = 1/\sqrt{3}, \quad b = \sqrt{3}$
- C $a = \sqrt{3}, \quad b = 1/\sqrt{3}$
- D $a = \sqrt{3}/2, \quad b = 2/\sqrt{3}$

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① $\sqrt{ab} = 1$

② $30° = 2\arctan\sqrt{\frac{b}{a}} - 90°$

$\hookrightarrow 60° = \arctan\left(\sqrt{\frac{b}{a}}\right) \rightarrow \sqrt{\frac{b}{a}} = \sqrt{3} \rightarrow \frac{b}{a} = 3$

$\rightarrow$ ①: $ab = 1 \rightsquigarrow a = 1/b$

in ②: $b^2 = 3 \rightsquigarrow b = \sqrt{3}, \ a = 1/\sqrt{3}$

Quiz 5

Problem: Let

$$P(s) = \frac{1}{s+1}, \text{ and } C(s) = \frac{k+s}{s},$$

respectively represent the transfer functions of a linear time-invariant system and corresponding PI-controller, where $k \in \mathbb{R}$ is a tuning parameter. Further, let P and C be interconnected according to the standard feedback architecture shown in Figure 15.

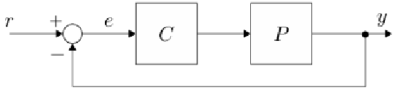

Figure 15: Standard feedback architecture. System P and controller C.

Q32 **(1.0 Points)** The tuning parameter k is to be chosen such that the gain crossover frequency ω_{gc} of the open-loop system is at 10 rad/s. Which choice of k satisfies the aforementioned requirement? Mark the correct answer.

- ☐ $400\,\pi^2$.
- ■ 100.
- ☐ $\frac{\pi}{20}$.
- ☐ 0.25.

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$$L(s) = C(s)P(s) = \frac{1}{s+1} \cdot \frac{k+s}{s}$$

$$= \frac{k+s}{s(s+1)} \qquad \omega_{gc} = 10 \text{ rad/s}$$

$$\rightarrow |L(j\omega_{gc})| \overset{!}{=} 1 \qquad L(j\omega_{gc}) = L(10j) = \frac{k+10j}{-100+10j}$$

$$\rightarrow \left|\frac{k+10j}{-100+10j}\right| \overset{!}{=} 1$$

$$|k+10j| = |-100+10j|$$

$$\Leftrightarrow k = 100$$

Course Schedule

	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	Polar plots, The Nyquist condition	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Compensators, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

Today

1. Time-delays
2. Designing Compensators
3. Non-Linearities

1. Time-delays

Introduction

Why?

Delays are common in control systems:

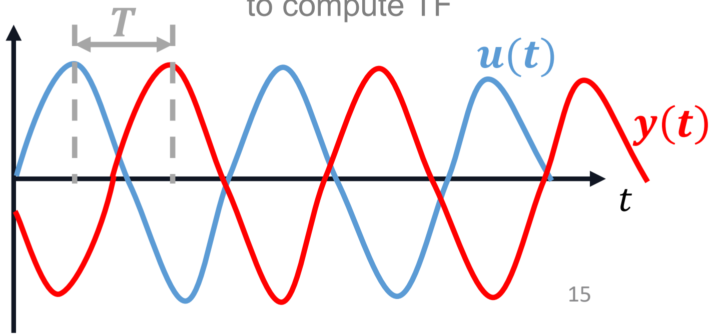
- If your controller is implemented on a computer, then reading sensor data and calculating the input for your actuator will take a finite computation time
- They can also be part of your physical plant

How?

- Time delay operator transforms input $u(t)$ into delay $y(t) = u(t-T)$, with $T \geq 0$
- We can compute TF by considering an input $u(t) = e^{st}$ ← like we always do to compute TF

$\longrightarrow$ the output will be $y(t) = e^{s(t-T)} = e^{-sT}u(t)$

$\longrightarrow$ hence $\boxed{G(s) = e^{-sT}}$



$$G(s) = e^{-sT}$$

Frequency Response of Time Delays

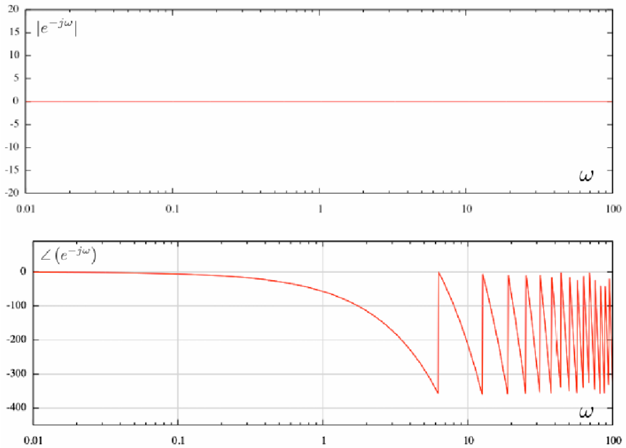
… this is NOT a rational transfer function, with which we have been dealing all this time

- No poles or zeros

Frequency response of a time delay : $G(j\omega) = e^{-j\omega T}$

- Magnitude $\left|e^{-j\omega \mathrm{T}}\right| = 1$
- Phase: $\angle e^{-j\omega T} = -\omega T$

Bode & Polar Plot:



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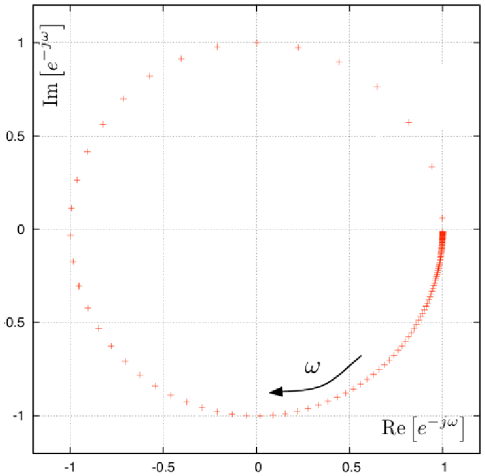
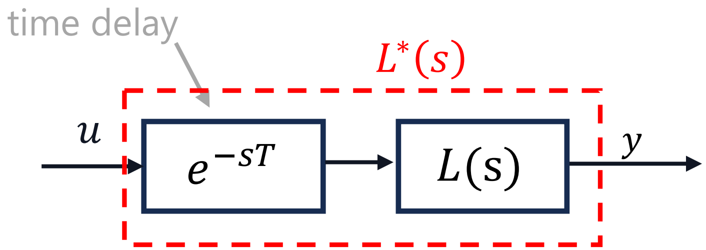


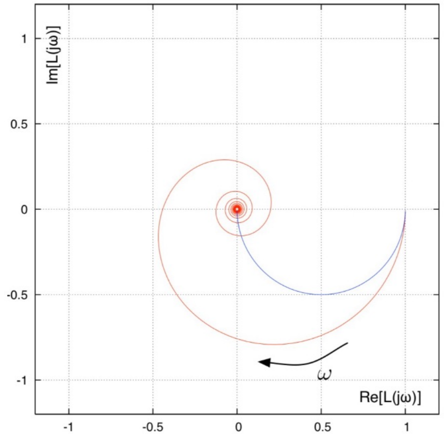
Effects of Time Delays



Now that we know the behavior of a pure delay, let's combine it with a system!

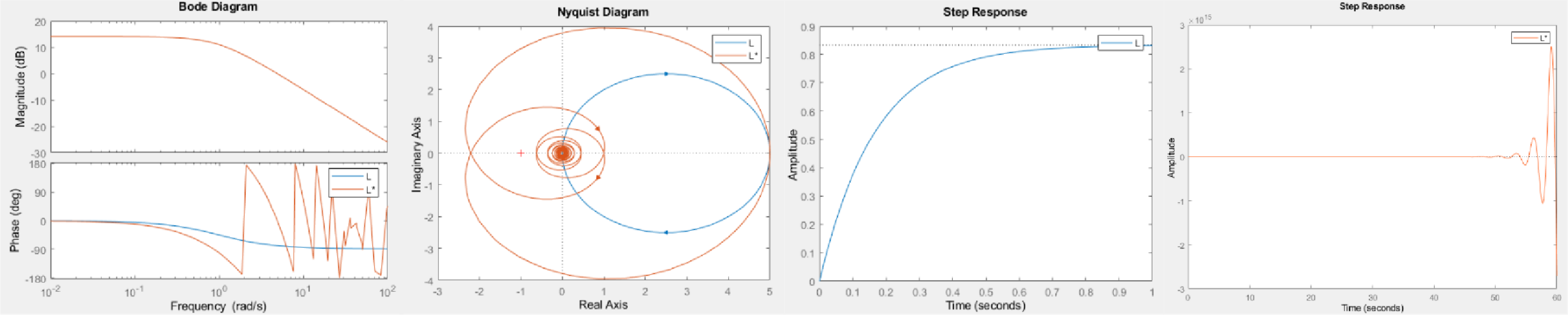
- $|L^*(s)| = |L(s)|$
- $\angle L^*(s) = \angle L(s) - \omega T$

- time delays cause the phase to drop by $\omega T \longrightarrow$ $\boxed{PM_{with\ delay} = PM_{without\ delay} - \omega_c T}$
- Nyquist plots of delayed systems have these encirclements



Effects of Time Delays

E.g. $L^{*}(s) = e^{-sT}\,\dfrac{k}{s+1}$



Design Procedure:

1. Design a feedback control system ignoring the time delay
2. Check the effective phase margin (phase margin of the system with delay)
3. Redesign the controller until a satisfactory controller is found (increase phase @ ω_c, decrease ω_c, etc.)

Rationalizing Time Delay TF

This is a very "duct-tape-engineering" approach... Is there another way to solve this? Yes!

- We can use some mathematical tricks to change the representation of a delay

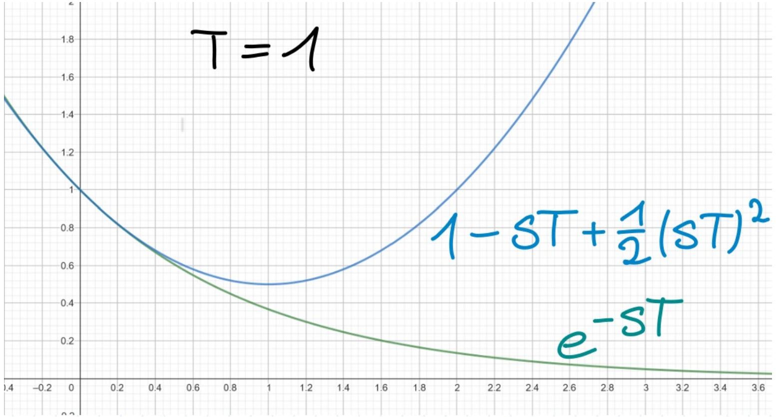
⇒ Technically we can write e^{-sT} as a polynomial! Using Taylor-approximation:

⇒ Can use for e.g. Root Locus!

$$e^{-sT} = 1 - sT + 1/2(sT)^2 - 1/6(sT)^3 + \cdots$$

An infinite series is hard to implement, therefore we approximate:

$$e^{-sT} \approx 1 - sT + 1/2(sT)^2$$



We note that the magnitude of the approximation diverges for $\omega \rightarrow \infty$, even though $\left|e^{-j\omega T}\right| = 1$

solid for small ω, but bad approximation for $\omega \longrightarrow \infty$

better approximation

Padé Approximation

For simplicity we use a ratio of first order polynomials: $e^{-sT} \approx k\frac{s+p}{s+q}$

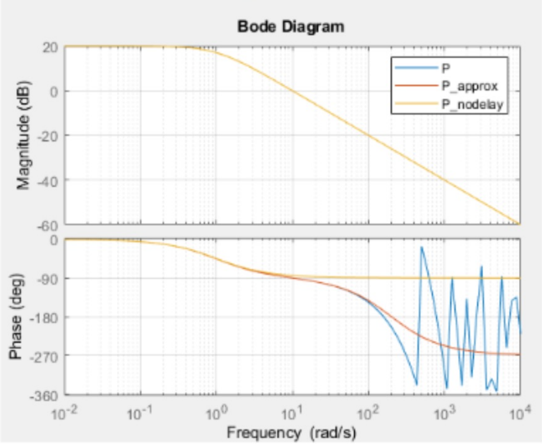
Setting the Padé Approximation equal to the Taylor expansion:

$$k \cdot \frac{s+p}{s+q} = 1 - sT + \frac{1}{2}(sT)^2 - \frac{1}{6}(sT)^3 + \cdots$$

Solving for the coefficients we get: $q = \frac{2}{T}, k = -1, p = -\frac{2}{T}$

Padé Approximation

$$\boxed{e^{-sT} \approx \frac{\frac{2}{T} - s}{\frac{2}{T} + s}}$$



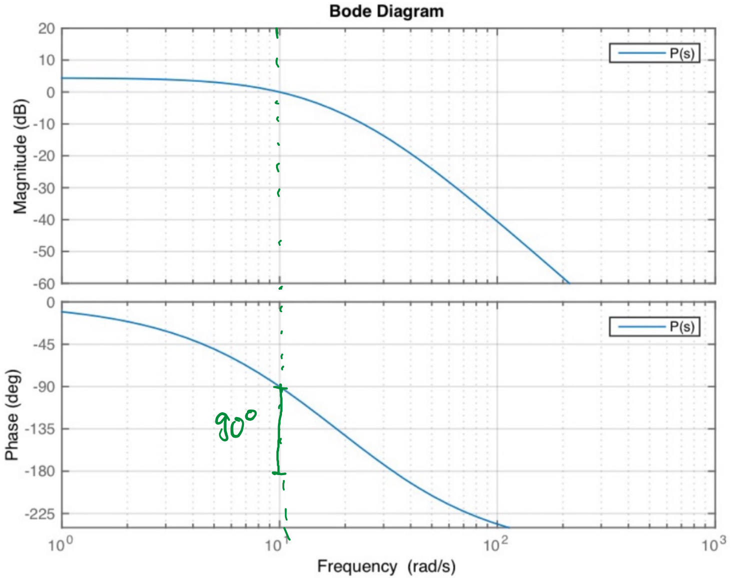
Note: The presence of the NMP zero indicates a limit for the gain k and with it a limit for the crossover frequency!
Keep in mind: This is an approximation, always use Nyquist to verify CL stability!

$$PM_{with\ delay} = PM_{without\ delay} - \omega_c T$$

Exam Problem

Question 45 *Choose the correct answer. (1 Point)*

You are given the Bode plot of a plant however the time delay has not been included in the model. You have a time delay of $T_d = \frac{\pi}{40}\ s$. What is the new phase margin/ phase reserve of your system?



a) $\varphi_{new} = 45°$

b) $\varphi_{new} = 30°$

c) $\varphi_{new} = 60°$

d) $\varphi_{new} = 90°$

$PM_{without\ delay} = 90°$

$PM_{with\ delay} = 90° - \frac{\pi}{40} \cdot 10 = 90° - \frac{\pi}{4}$

$= 45°$

$$PM_{with\ delay} = PM_{without\ delay} - \omega_c T$$

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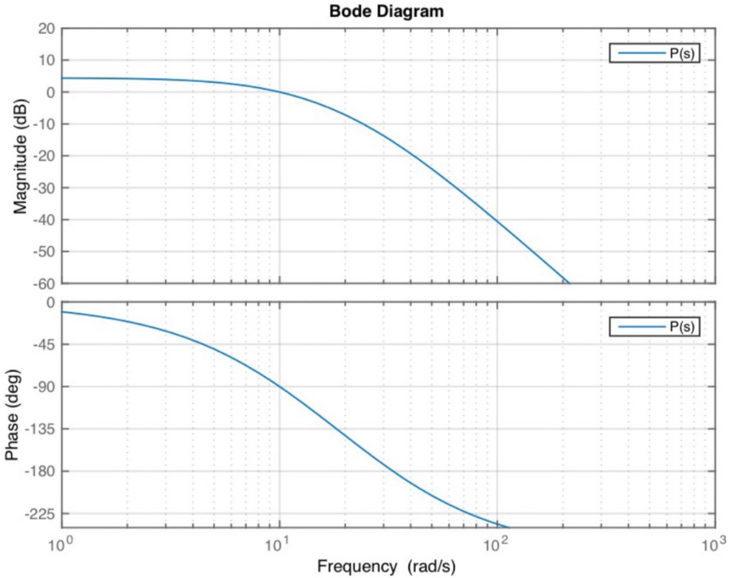

a) $\varphi_{new} = 45°$

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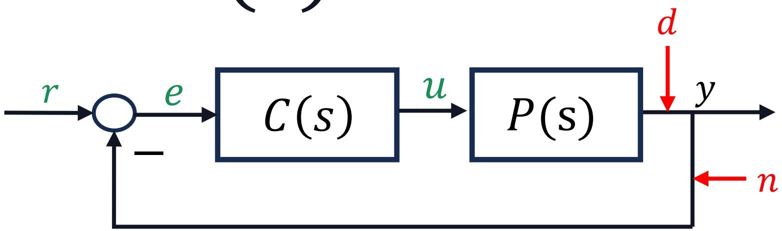
2. Designing Compensators

Implementing a Controller $C(s)$

- Controller a.k.a compensator

→ implemented on a computer



Recall: *Transfer Function*

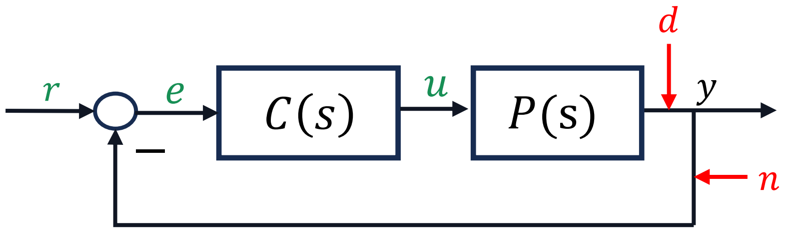
$$C(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0}$$

Input: $u = e = r - y$

$$\dot{x}(t) = Ax(t) + Be(t)$$
$$y(t) = Cx(t) + De(t)$$

$$\left[\begin{array}{c|c} A & b \\ \hline c & d \end{array}\right] = \left[\begin{array}{ccccc|c} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & & 1 & 0 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} & 1 \\ \hline b_0 & b_1 & b_2 & \cdots & b_{n-1} & d \end{array}\right]$$

Implementing a Controller $C(s)$

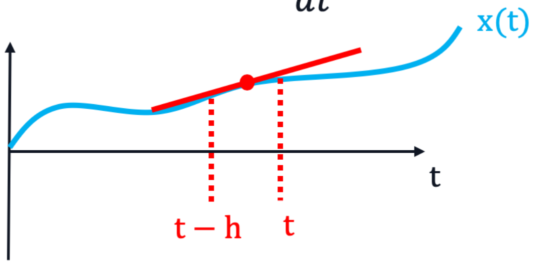


Discretize a continuous signal

→the smaller dt the more accurate the approximation of the signal

$$\dot{x}(t) = \lim_{h \to 0} \frac{x(t) - x(t-h)}{h}$$

$$\dot{x}(t) \approx \frac{x(t) - x(t-dt)}{dt}$$



Initialize the state to, e.g., $x \leftarrow 0$.
Let dt be some small time interval.
repeat (every dt seconds)
- Read the reference value r.
- Read the measured output value y.
- Compute the error e.
- Update the state as $x \leftarrow x + (Ax + Be) * dt$.
- Compute the output as $u = Cx + De$.
- Send the command u to the actuators.

until Forever

3. Non-Linearities

Non-Linearities

- Tools we have learnt in CS1 only valid for linear systems
- Most systems in real world are non-linear
- to bridge this gap → linearize systems around equilibrium points
- we did this with Jacobian linearization procedure that takes some non-linear system:

$$\dot{x}(t) = f\big(x(t), u(t)\big)$$
$$y(t) = h\big(x(t), u(t)\big)$$

and produces a linearized system valid around the equilibrium point. The system is given by:

$$A = \frac{\partial f(x,u)}{\partial x}\bigg|_{(x_e,u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}\Bigg|_{(x_e,u_e)} \in \mathbb{R}^{n\times n} \qquad C = \frac{\partial g(x,u)}{\partial x}\bigg|_{(x_e,u_e)} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \cdots & \frac{\partial g_n}{\partial x_n} \end{bmatrix}\Bigg|_{(x_e,u_e)} \in \mathbb{R}^{p\times n}$$

$$B = \frac{\partial f(x,u)}{\partial u}\bigg|_{(x_e,u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}\Bigg|_{(x_e,u_e)} \in \mathbb{R}^{n\times m} \qquad D = \frac{\partial g(x,u)}{\partial u}\bigg|_{(x_e,u_e)} = \begin{bmatrix} \frac{\partial g_1}{\partial u_1} & \cdots & \frac{\partial g_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial u_1} & \cdots & \frac{\partial g_n}{\partial u_m} \end{bmatrix}\Bigg|_{(x_e,u_e)} \in \mathbb{R}^{p\times m}$$

Hartman–Grobman

Given this linearization, the Hartman–Grobman theorem tells us that if the linearized system is closed-loop BIBO stable, then the nonlinear system is also stable in a range around the equilibrium point. We just don't know how large that region is.

If we now want to design a controller for a nonlinear system we proceed as follows:

1. Design a linear compensator for the linear model
2. If the system is CL stable the nonlinear system will also be stable in the region of the equilibrium
3. Check in a nonlinear simulation if your design is robust with respect to typical deviations.

Appendix

$$A = \left.\frac{\partial f(x,u)}{\partial x}\right|_{(x_e,u_e)} = \left.\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}\right|_{(x_e,u_e)} \in \mathbb{R}^{n\times n}$$

Question 4 *Choose the correct answer. (1 Point)*

Consider the Van der Pol oscillator with differential equation

$$\ddot{q} + (q^2 - 1)\dot{q} + q = 0.$$

For a state-space representation define $x_1 := q$ and $x_2 := \dot{q}$.
You are given the equilibrium point $x_e = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
Linearize the system with respect to the equilibrium.
What is the resulting dynamics matrix A?

A $A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$

B $A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$

C $A = \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}$

D $A = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$

$x_1 = q \quad x_2 = \dot{q}$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ -q - \dot{q}(q^2-1) \end{bmatrix}$$

$$\hookrightarrow \dot{x} = \begin{bmatrix} x_2 \\ -x_1 - x_2(x_1^2 - 1) \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -(x_1^2 - 1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$$

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Tips for Problem Set 11

Easy	Medium	Hard
	1	

1.1) recall gain & phase margins from last week & how to close the Nyquist plot if there are poles on the imaginary axis

1.2) recall formula for Padé Approximation & plot Root Locus, Bode Plot & Nyquist with the tools (or look up in solutions) . For Bode Plot we need positive margins for stability and for Nyquist Plot recall Nyquist Criterion

1.3a) look up lead/lag-elements from last week, lag elements introduce phase lag.

For the condition $\frac{a}{b} = 10$: Influence of disturbance on error is given by $|S(j\omega)| = \frac{1}{|1+L(j\omega)|} \approx \frac{1}{|L(j\omega)|}$, so for an error smaller than 1% we need $|L(j\omega)| > 100$. The current magnitude at 0.01 rad/s is: $|L(0.01j)| \approx 10$, so we need an increase in magnitude of 10.

1.3b) which element increases phase? what is the current phase margin and how much do we need to increase it? use the following formula: $\varphi_{\max} = 2\arctan\left(\sqrt{\frac{b}{a}}\right) - 90°$

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.

https://docs.google.com/forms/d/e/1FAIpQLSdHl0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header